

Deep learning avec Pytorch

Partie 4 : La régression linéaire



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Problème de régression

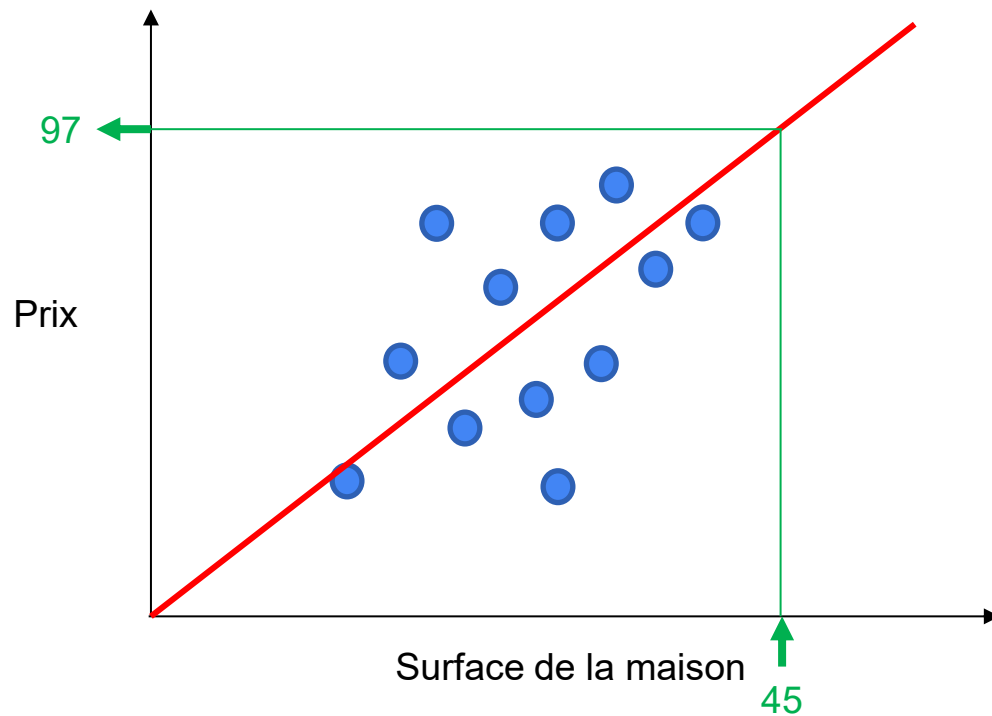


	Surface (x_1)	Nb de pièces(x_2)	Année (x_3)	Prix (y)
1	70	3	2010	460
2	40	3	2015	232
3	45	4	1990	315
4	12	2	2017	178
...	...	...	...	...
m	25	1	2005	240

Jeu d'entraînement pour la prédiction de prix de maison

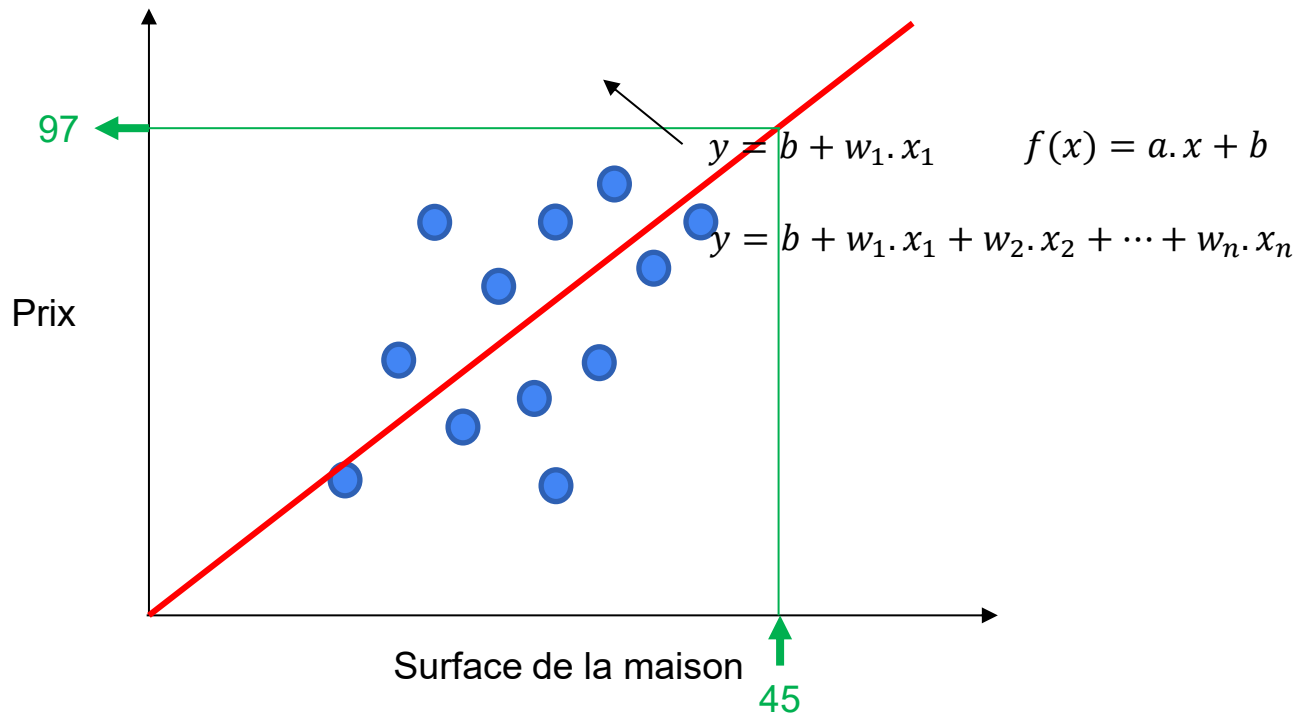


Utiliser une droite





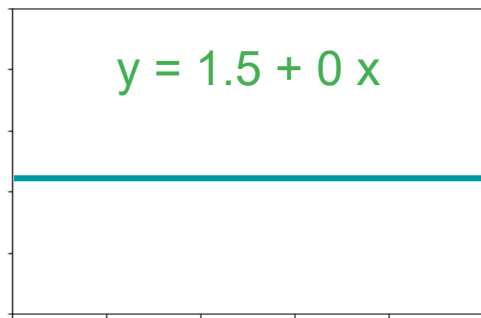
Equation d'une droite





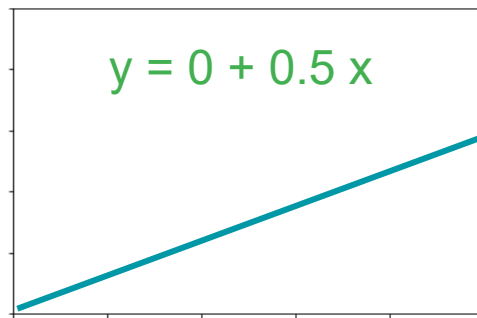
Les paramètres de la régression linéaire

$$y = b + w_1 x$$



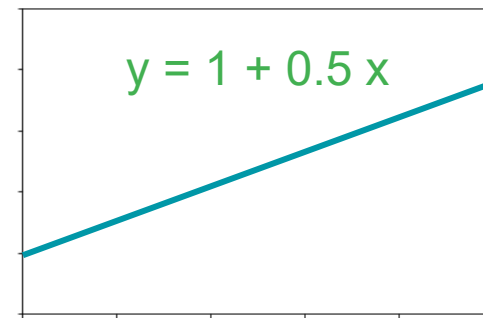
$$b = 1.5$$

$$w_1 = 0$$



$$b = 0$$

$$w_1 = 0.5$$

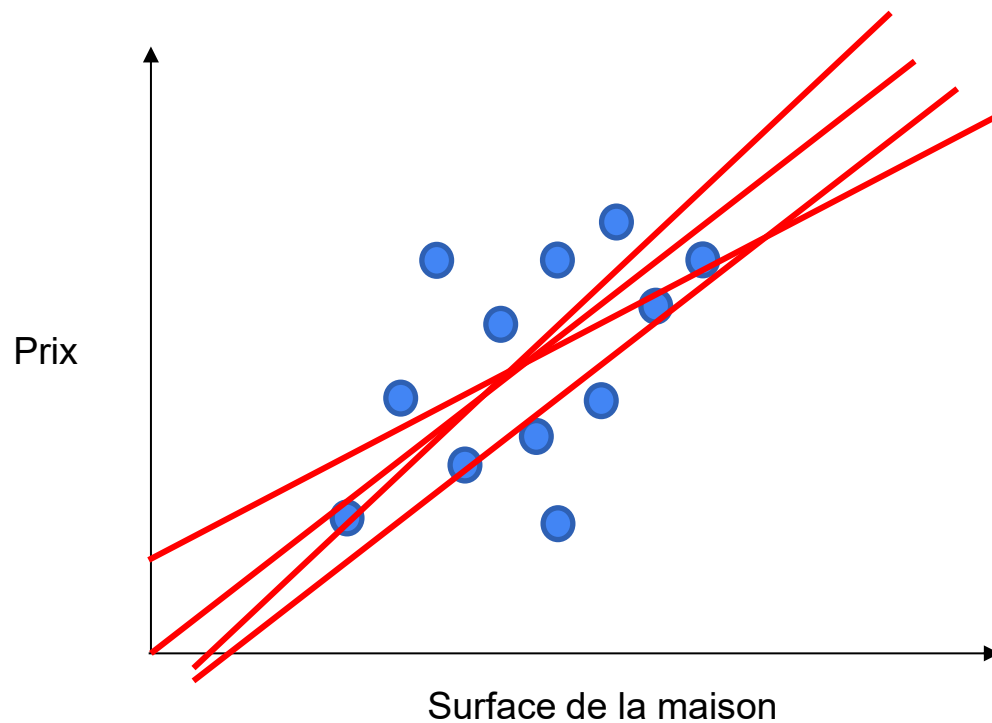


$$b = 1$$

$$w_1 = 0.5$$

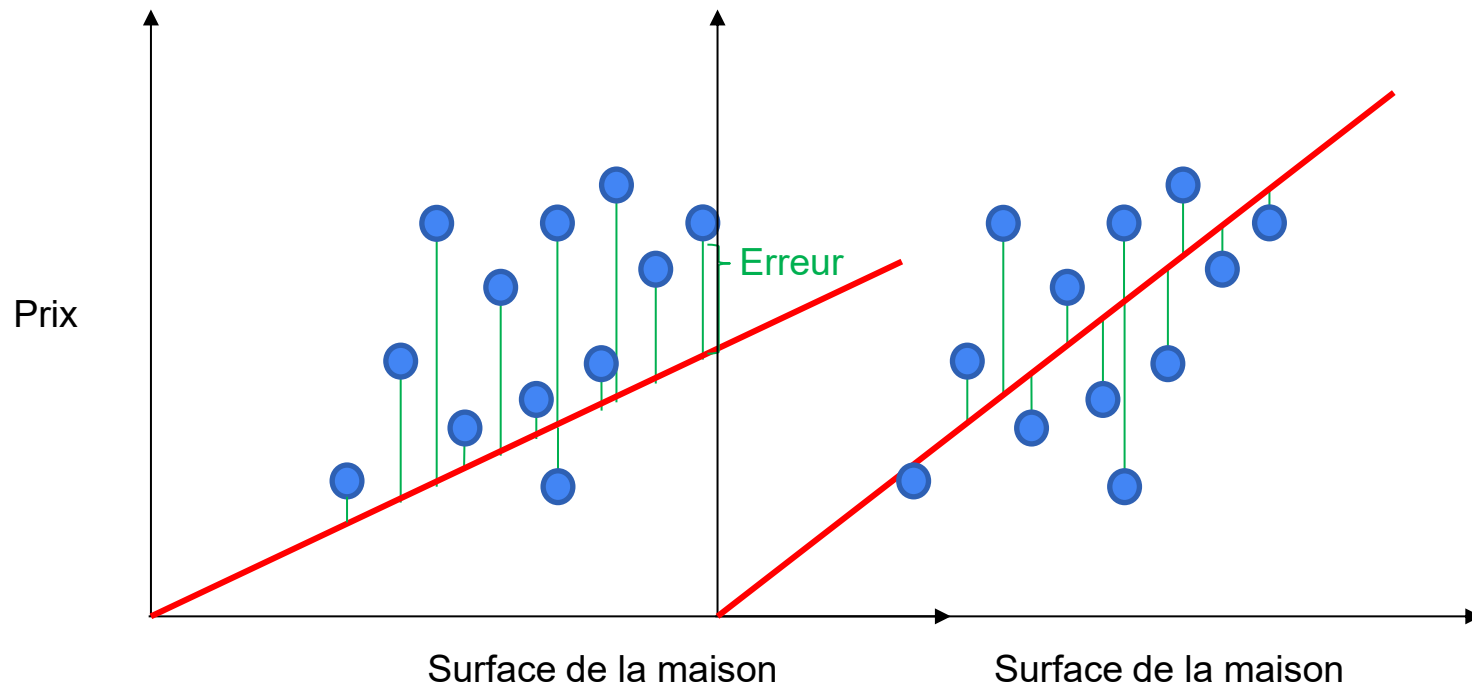


Quels paramètres choisir ?





La notion d'erreur





L'erreur moyenne au carré

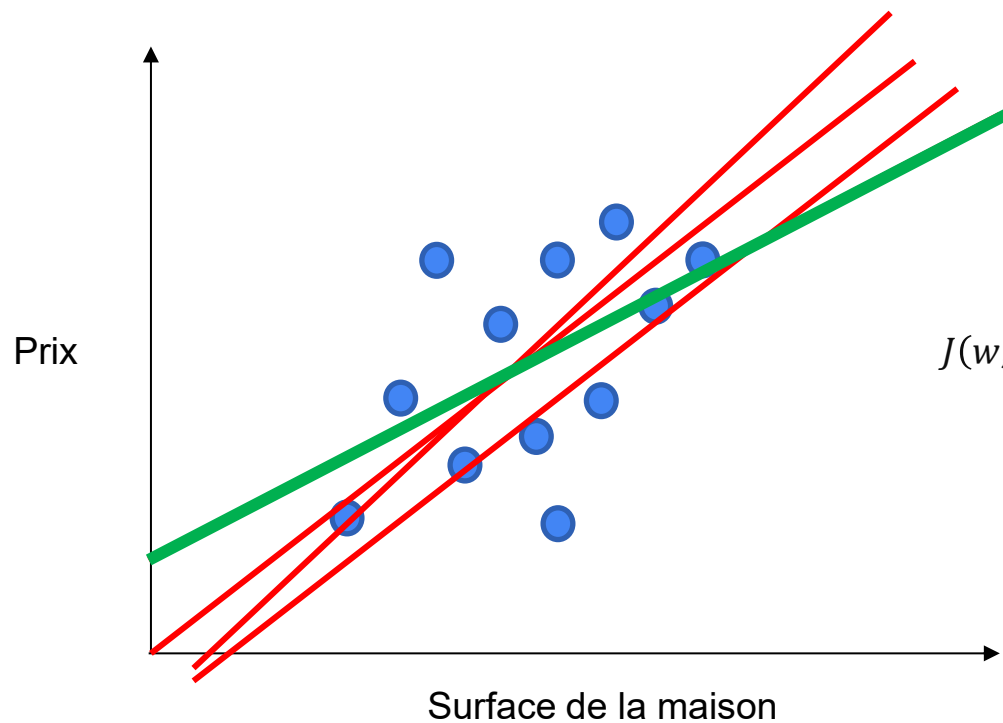
$$\begin{array}{cccc} \hat{y} - y & \hat{y}_1 = 500 & y_1 = 520 & \hat{y}_1 - y_1 = -20 \\ \frac{1}{m} \sum \hat{y} - y & \hat{y}_2 = 350 & y_2 = 320 & \hat{y}_1 - y_1 = 30 \end{array}$$

$$\frac{1}{2} \sum_{i=1}^2 \hat{y}_{(i)} - y_{(i)} = \frac{10}{2} = 5$$

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_{(i)} - y_{(i)})^2$$



Quels paramètres choisir ?



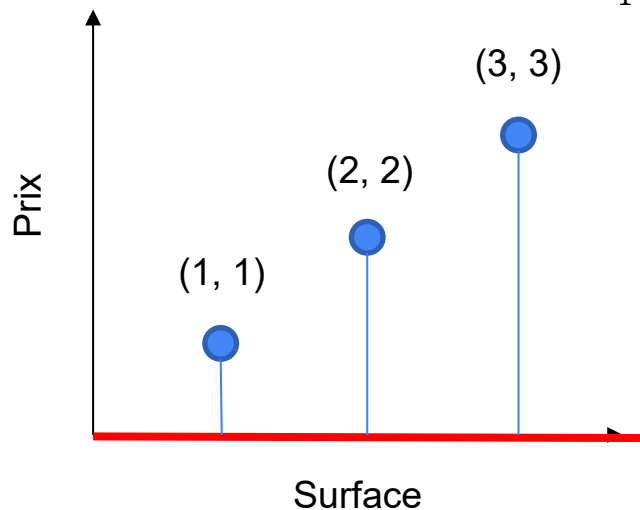
$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_{(i)} - y_{(i)})^2$$



Tracer la fonction de coût

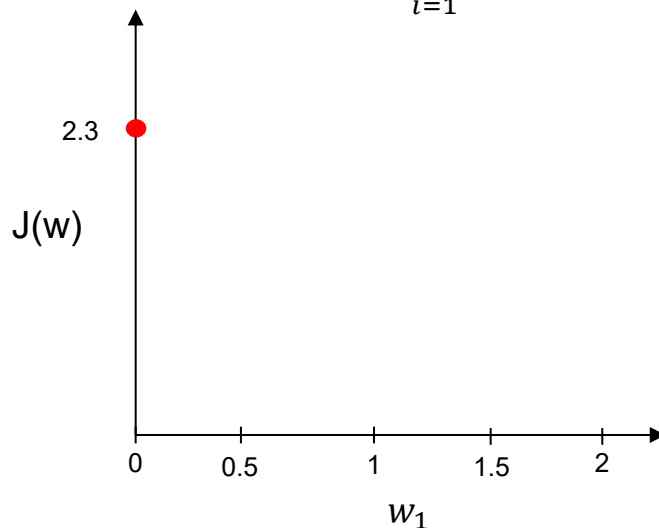
$$\hat{y} = b + w_1 \cdot x_1$$

b est fixé à 0
 w_1 va varier



$$\hat{y} = 0 + 0 \cdot x_1$$

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_{(i)} - y_{(i)})^2$$



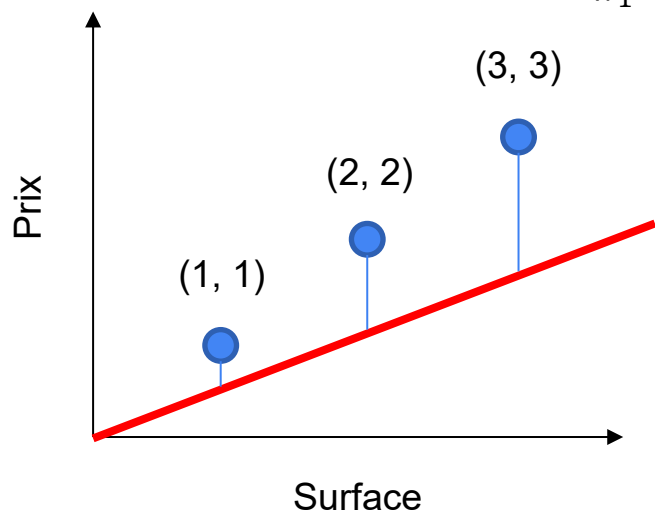
$$J(0) = \frac{1}{2m} [(1)^2 + (2)^2 + (3)^2] = 2.3$$



Tracer la fonction de coût

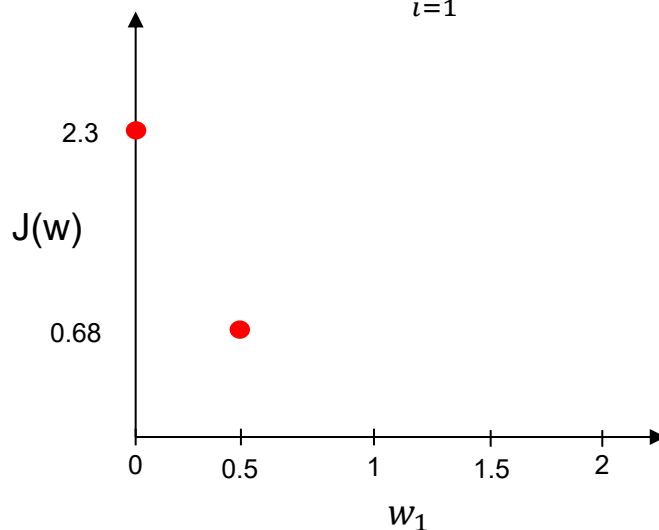
$$\hat{y} = b + w_1 \cdot x_1$$

b est fixé à 0
 w_1 va varier



$$\hat{y} = 0 + 0.5 \cdot x_1$$

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_{(i)} - y_{(i)})^2$$



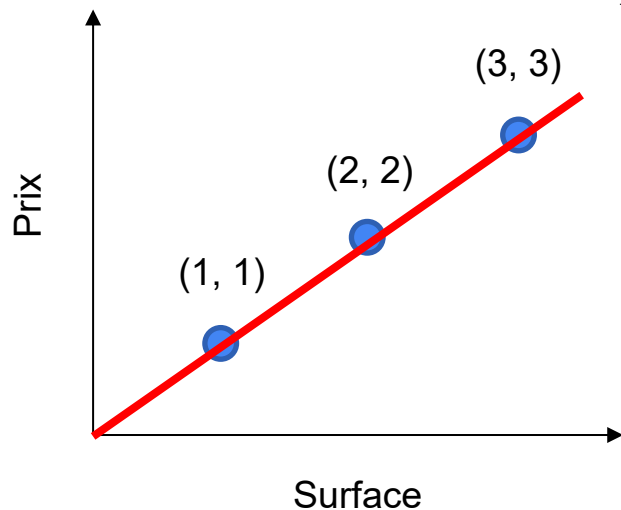
$$J(0.5) = \frac{1}{2m} [(0.5 - 1)^2 + (1 - 2)^2 + (1.5 - 3)^2] = 0.68$$



Tracer la fonction de coût

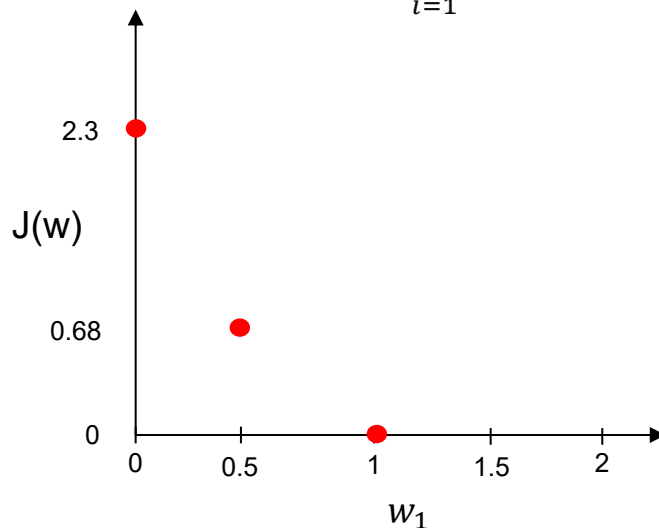
$$\hat{y} = b + w_1 \cdot x_1$$

b est fixé à 0
 w_1 va varier



$$\hat{y} = 0 + 1 \cdot x_1$$

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_{(i)} - y_{(i)})^2$$



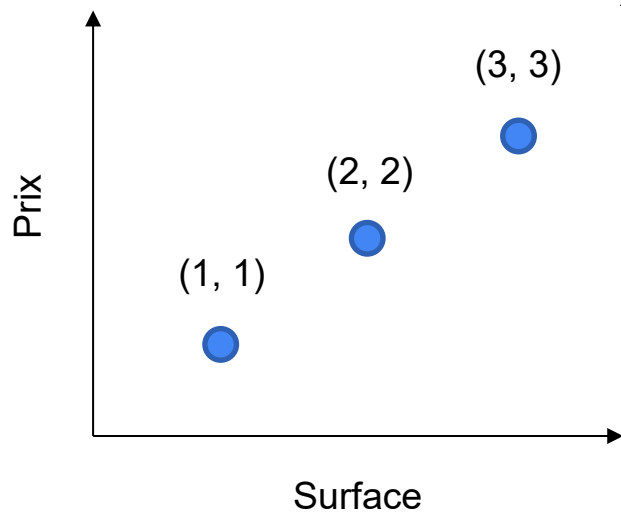
$$J(1) = \frac{1}{2m} [(1 - 1)^2 + (2 - 2)^2 + (3 - 3)^2] = 0$$



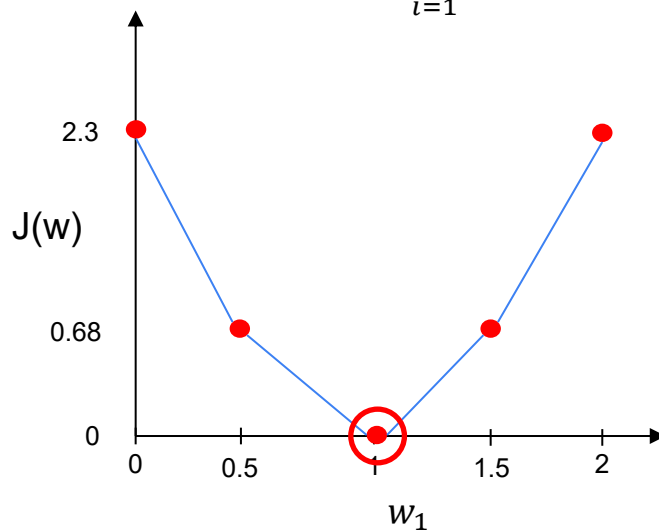
Tracer la fonction de coût

$$\hat{y} = b + w_1 \cdot x_1$$

b est fixé à 0
 w_1 va varier

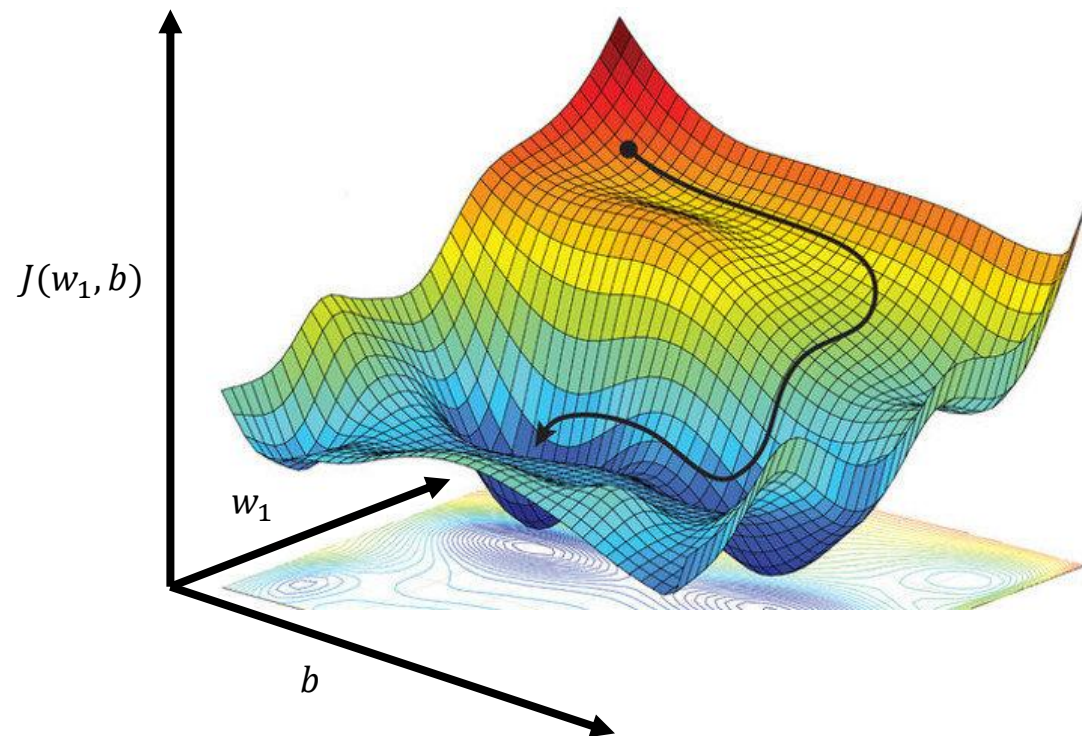


$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_{(i)} - y_{(i)})^2$$





Fonction de coût avec deux paramètres





Le gradient descent

Soit n le nombre de variables

Répéter ce processus jusqu'à la convergence :

$$w_j := w_j - \alpha \frac{\partial}{\partial w_j} J(w, b) \quad j \in [1, n]$$

Nouvelle valeur de w_j

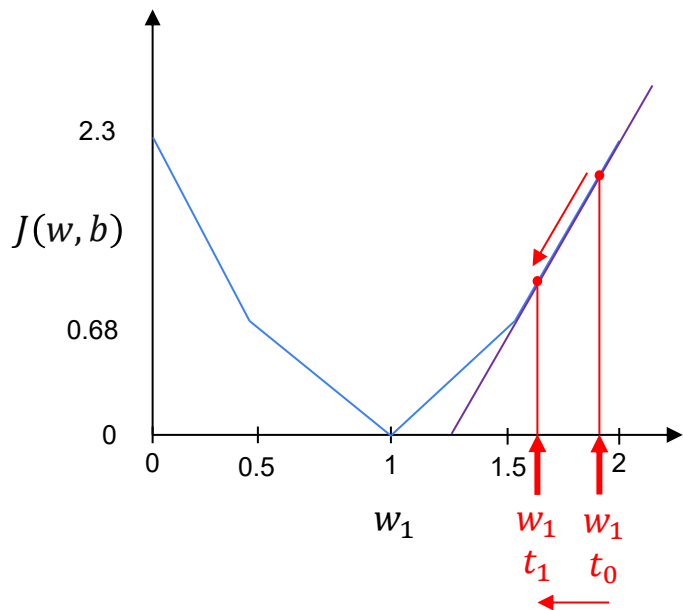
Ancienne valeur de w_j

Le learning rate

Dérivée partielle



La dérivée partielle



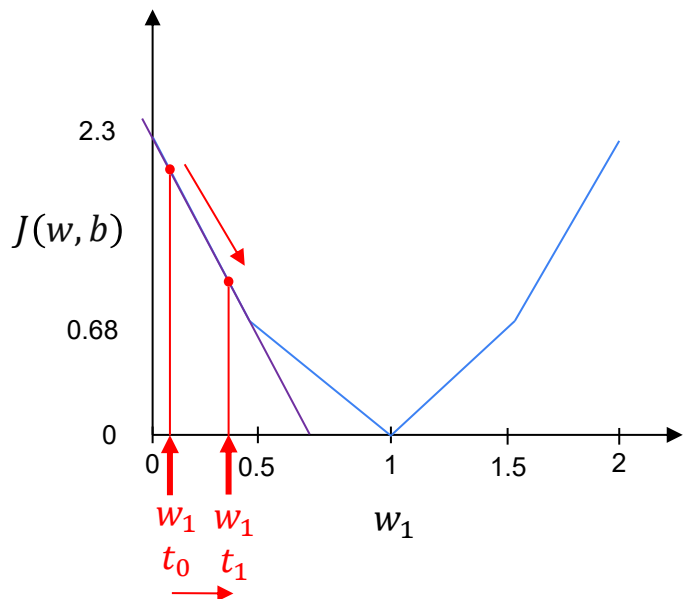
$$w_1 := w_1 - \alpha \frac{\partial}{\partial w_1} J(w, b)$$
$$t_1 \quad t_0$$

$$w_1 := w_1 - \text{positif}$$
$$t_1 \quad t_0$$

$$w_1 < w_1$$
$$t_1 \quad t_0$$



La dérivée partielle



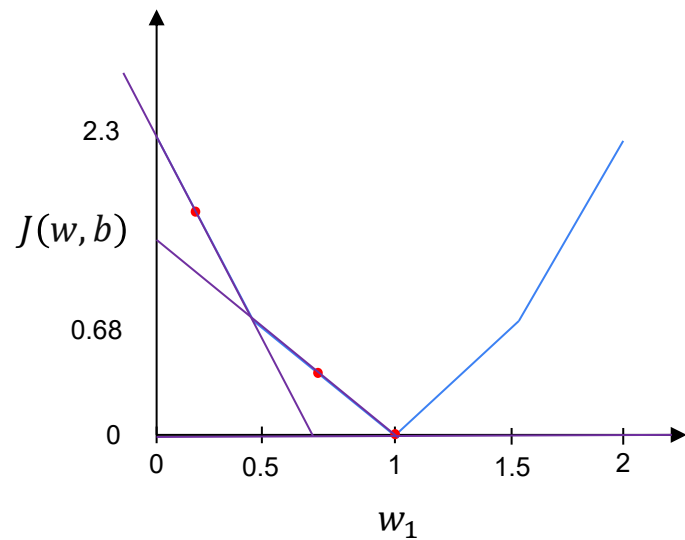
$$w_{1,t_1} := w_{1,t_0} - \alpha \frac{\partial}{\partial w_1} J(w, b)$$

$$w_{1,t_1} := w_{1,t_0} - \text{negatif}$$

$$w_{1,t_1} > w_{1,t_0}$$



Atteindre le minimum



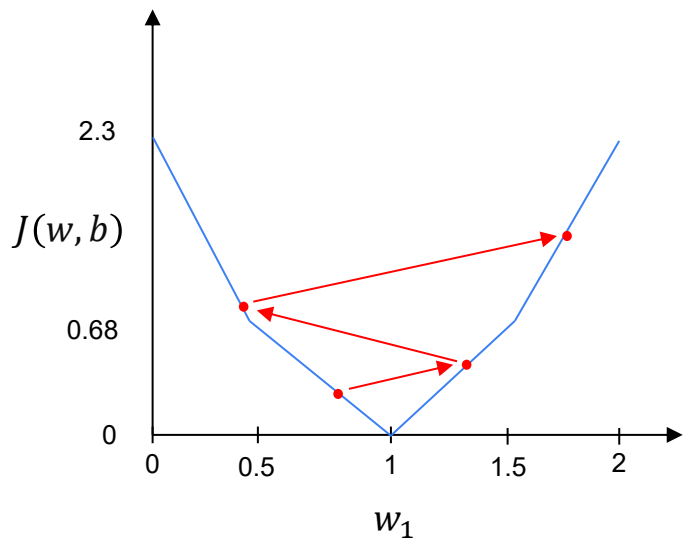
Réduit vers le minimum

$$w_1 := w_1 - \alpha \frac{\partial}{\partial w_1} J(w, b)$$

Est fixe



Impact du learning rate

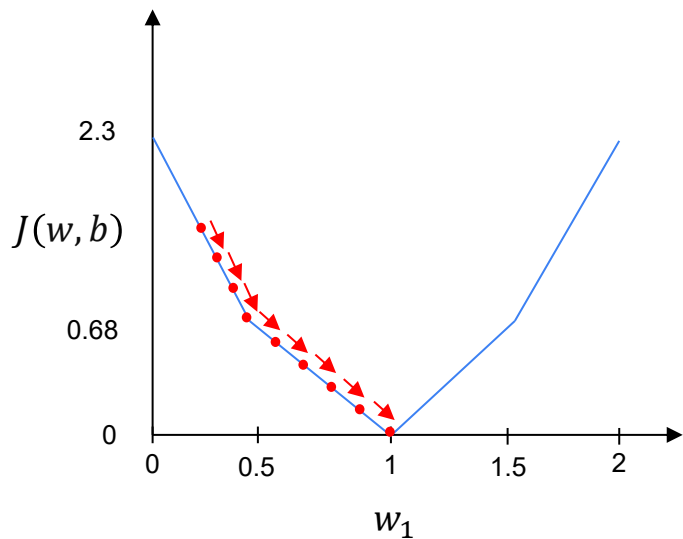


$$w_1 := w_1 - \boxed{\alpha} \frac{\partial}{\partial w_1} J(w, b)$$

Learning rate trop grand



Impact du learning rate



$$w_1 := w_1 - \boxed{\alpha} \frac{\partial}{\partial w_1} J(w, b)$$

Learning rate trop petit